Field Survey Protocols

Vegetation Survey Methods

The study will survey approximately 20 neutral grassland located throughout Worcestershire of varying sizes and age classes[1], including both known ancient and recent grasslands. Site areas range from 0.25 ha to a maximum 30 ha, reflecting real-world variability in habitat extent.

To account for the well-established relationship between species richness and sampling area, a logarithmic sampling intensity approach will be used. This approach balances representativeness, sampling efficiency, and ecological comparability across sites of different sizes.

Quadrat Allocation Method

Each site will be surveyed using the following formula to determine the number of sampling quadrats:

$$\text{Quadrats per site} = \max\left(10, \left[10 + \log 10\ \left(\frac{Site\ area\ (ha)}{0.1}\right) \times 10\right]\right)$$

This ensures a minimum of 10 quadrats per site which prevents under sampling in smaller areas and ensures an increasing number of quadrats with site size, scaled logarithmically to reflect diminishing returns in species accumulation with area and reducing trampling impacts by increasing efficiency. This equates to 10 – 15 quadrats per field compartment.

Quadrat Placement and Data Collection

Quadrats (2 m^2) will be placed using a stratified-random approach, ensuring coverage of the site's environmental heterogeneity (e.g., microtopography, management zones, visible gradients). Within each quadrat, the following data will be collected:

- Grid Reference (SW corner of quadrat)
- Percentage cover of all vascular plants
- Percentage cover of all bryophytes
- Percentage cover of all microlichens (where readily identifiable)
- Percentage cover of bare ground
- Mean vegetation height (x5 readings in each corner of quadrat and centre)

[1] Where 'age class' is defined by years of known continuity of management / absence of significant improvement.

- Slope
- Aspect
- Number of anthills
- Number of mole hills
- Soil texture

Quadrats will always be orientated north, and the SW corner will be recorded with GPS to allow for replication if ever needed. It is accepted that this cannot be 100% accurate.

In addition to quadrat derived data, a walk over of the site, inclusive of atypical regions, boundaries and areas with different management treatments will be conducted to collect a full site species list. Species will be assigned DAFOR values (Dominant, Abundant, Frequent, Occasional, Rare)[2]. This data will not be subject to rigorous statistical analysis but serves as a way of understanding general trends and for identifying key species that may have been missed in quadrat sampling. E.g., indicators of continuity associated with shaded plots, wetter areas and hedge banks.

Sampling for Genetic Analysis Field Methods

Sample Collection

Where possible, leaf samples (either partial or entire dependant on the organism) will be collected from individuals spread evenly across the site and at least 5 m apart by collecting one sample from each corner of a 5 m X 5 m quadrat, up to 30 leaf samples will be collected from each site where viable and only where deemed to not have deleterious impacts on the population(s) at a site.

Population Size Estimation

Estimations of population size will be made according to the following categories: <100 small; 100-1000 medium; >1000 large, though this may be adjusted depending on the individual species selected for analysis as what constitutes a large population between species may differ.

Soil Analysis Field Methods

Soil sampling will take place between February and April 2026, before plant growth resumes, to minimise trampling and disturbance. Each site will be sampled for pH, macronutrients (P, K, Mg), organic matter, and microbial activity.

[2] D – Dominant (50 – 100%); A – Abundant (30 – 50%), F – Frequent (15 – 30%), O – Occasional (5 – 15%), R – Rare (<5%)

Soil Sampling Design

5–10 random sampling points will be selected per site depending on site size and variability. Points will be spread across site sectors using a stratified-random approach, avoiding field edges, dung patches, or visibly disturbed zones.

Soil Sampling Method

- Remove surface litter and extract soil from the top 10 cm using a soil corer or auger.
- Combine 5 cores into a composite sample per site (~500g) for pH and nutrient analysis.
- Collect a separate sterile subsample for microbial analysis.
- Label all bags clearly with site code, date, and sample point.

Optional field notes will include vegetation structure, recent management, soil moisture, and microhabitat context for each point.

All sampling is non-destructive and undertaken with landowner consent.